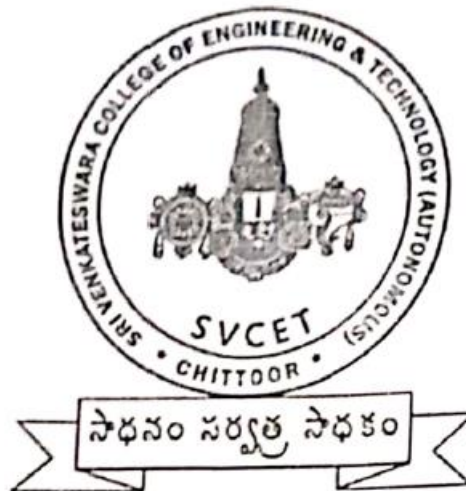


SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous)



NAME OF THE STUDENT :

ROLL NUMBER :

BRANCH & YEAR : CSD-C or CSM-C

SUBJECT WITH CODE : BASIC ELECTRICAL & ELECTRONICS

TOPIC : DC CIRCUITS & AC CIRCUITS

SUBMITTED ON :

SUBMITTED TO :

1) Define ohms law and its limitations and also KCL ,KVL?

Ans :

Ohms law : states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

$$v \propto i$$

Ohm defined the constant of proportionality for a resistor to be the resistance, R . (The resistance is a material property which can change if the internal or external conditions of the element are altered, e.g., if there are changes in the temperature.)

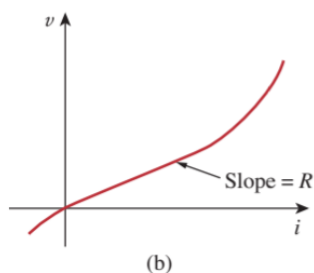
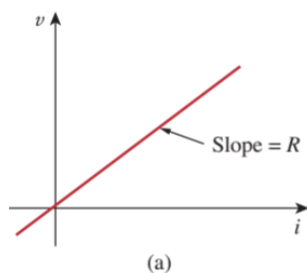
$$v = iR$$

Voltage :volts

Current :Ampere

Resistance :ohms

Limitations of ohms law:



a)Linear resistor b)non linear resistor

- 1) The non linear resistor does not obey ohms law eg : diode
- 2) Circuits varying with temperature not obey ohms law

Example : An electric iron draws 2 A at 120 V. Find its resistance

From Ohm's law,

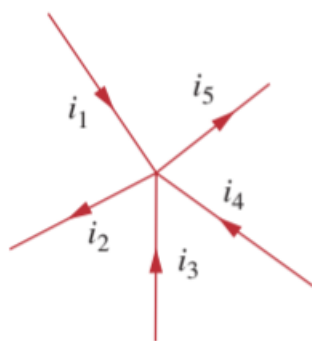
$$R = \frac{v}{i} = \frac{120}{2} = 60 \Omega$$

Kirchoffs laws

a)Kirchoff first law:

Kirchoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero

$$\sum_{n=1}^N i_n = 0$$



since currents i_1 , i_3 , and i_4 are entering the node, while currents i_2 and i_5 are leaving it. By rearranging the terms, we get

$$i_1 + i_3 + i_4 = i_2 + i_5 \quad (2.17)$$

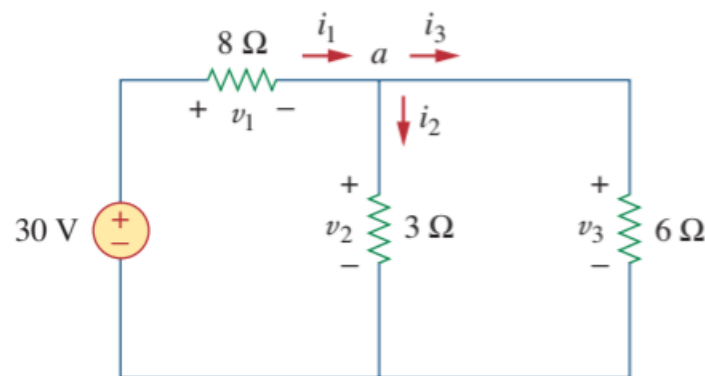
Kirchoff voltage law :

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

$$\sum_{m=1}^M v_m = 0$$

$$v_2 + v_3 + v_5 = v_1 + v_4$$

Find currents and voltages



(a)

Apply kcl $i_1 = i_2 + i_3$

Apply KVL to loop1 and loop2

$$30 = V_1 + V_2$$

$$V_3 - V_2 = 0$$

$$V_1 = 8 \cdot i_1 \quad ; \quad V_2 = 3 \cdot i_2 \quad ; \quad V_3 = 6 \cdot i_3$$

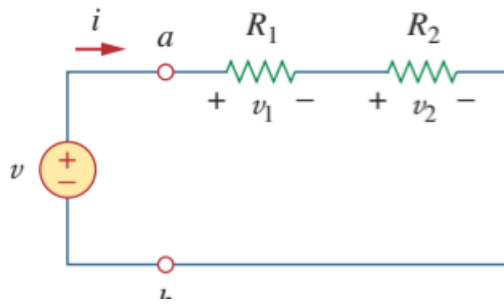
Solving we get

$$i_1 = 3A \quad ; \quad i_3 = 1A$$

$$V_1 = 24V \quad ; \quad V_2 = 6V \quad ; \quad V_3 = 6V$$

2') Explain series and parallel resistors with an example ?

Ans:



$$V_1 = i \cdot R_1 : v_2 = i \cdot R_2$$

Apply kvl

$$V = V_1 + V_2 = i(R_1 + R_2)$$

$$I = V / (R_1 + R_2)$$

$$V = i \cdot R_{eq}$$

$$R_{eq} = R_1 + R_2$$

The equivalent resistance of any number of resistors connected in series is the sum of the individual resistances

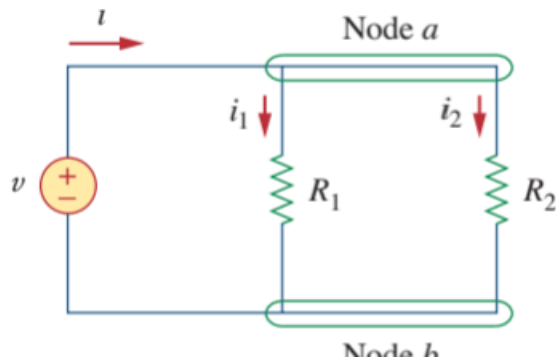
For N resistors in series then,

$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n$$

To determine the voltage across each resistor

$$v_1 = \frac{R_1}{R_1 + R_2} v, \quad v_2 = \frac{R_2}{R_1 + R_2} v$$

B) Parallel resistors:



$$I = I_1 + I_2$$

$$v/R_{eq} = v/R_1 + v/R_2$$

$$1/R_{eq} = 1/R_1 + 1/R_2$$

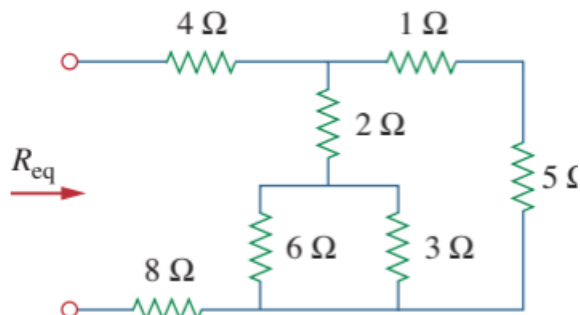
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

The equivalent resistance of two parallel resistors is equal to the product of their resistances divided by their sum.

For n resistances

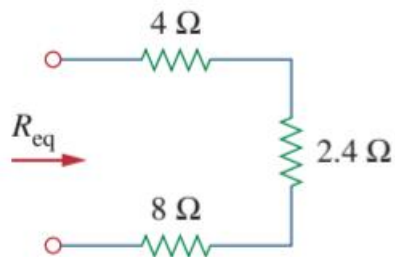
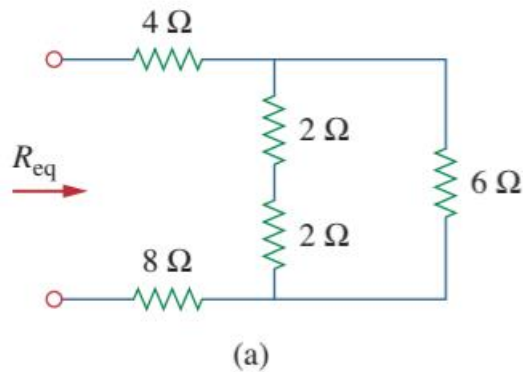
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

Example:\



$$6//3 = 6 \times 3 / (6 + 3) = 2\text{ohms}$$

1ohm and 5 ohm are in series = 1+5 =6ohms



This 4-Ω resistor is now in parallel with the 6-Ω resistor in their equivalent resistance is

$$4\ \Omega \parallel 6\ \Omega = \frac{4 \times 6}{4 + 6} = 2.4\ \Omega$$

the three resistors are in series. Hence, the equivalent resistance for the circuit is

$$R_{eq} = 4\ \Omega + 2.4\ \Omega + 8\ \Omega = 14.4\ \Omega$$

3) Explain superposition theorem with an example

Ans:

The idea of superposition rests on the linearity property.

The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

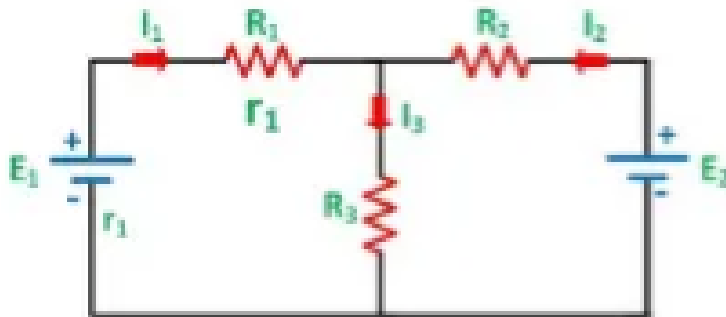
With these in mind, we apply the superposition principle in three steps:

Steps to Apply Superposition Principle: 1. Turn off all independent sources except one source. Find the

output (voltage or current) due to that active source using the techniques covered in Chapters 2 and 3.

2. Repeat step 1 for each of the other independent sources. 3. Find the total contribution by adding algebraically all the contributions due to the independent sources

Example :



$E_1 = 28\text{V}$ and $E_2 = 7\text{V}$

$R_1 = 4\ \Omega$ and $R_3 = 2\ \Omega$ and $R_2 = 1\ \Omega$

Apply nodal analysis

$$I_1 = I_3 + I_2$$

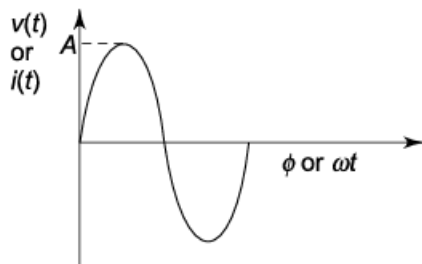
$$\frac{V - 28}{4} = \frac{V}{2} + \frac{V - 7}{1}$$

$$V = 8\text{V}$$

$$I_1 = 5\text{A} : I_3 = 4\text{A}, I_2 = 1\text{A}$$

4) Explain Average value ,RMS value,form factor ,peak factor , and power factor

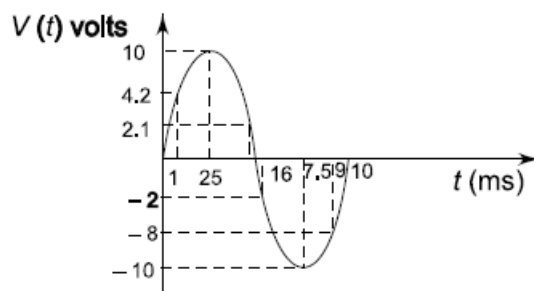
Ans:



A sine wave is graphically represented as shown in Fig. In general, the sine wave is represented by the equation

$$v(t) = V_m \sin \omega t$$

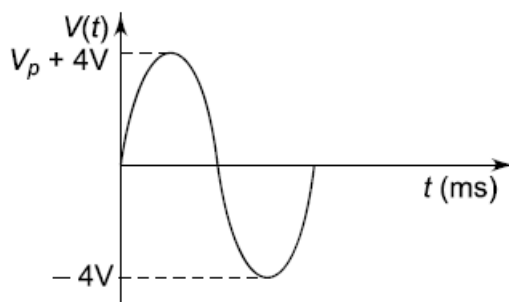
Instantaneous Value



shown at time 1 ms, the value is 4.2 V; the value is 10 V at 2.5 ms, -2 V at 6 ms and -10 V at 7.5 and so on.

Peak value :

Fig. 4.12



The peak value of the sine wave is the maximum value of the wave during positive half cycle, or maximum value of wave during negative half cycle. Since the value of these two are equal in magnitude, a sine wave is characterised by a single peak value. The peak value of the sine wave is **4V**

Average value :

In general, the average value of any function $v(t)$, with period T is given by

$$v_{av} = \frac{1}{T} \int_0^T v(t) dt$$

$$\begin{aligned} v_{av} &= \frac{1}{\pi} \int_0^{\pi} V_P \sin \omega t d(\omega t) \\ &= \frac{1}{\pi} [-V_P \cos \omega t]_0^{\pi} \\ &= \frac{2V_P}{\pi} = 0.637 V_P \end{aligned}$$

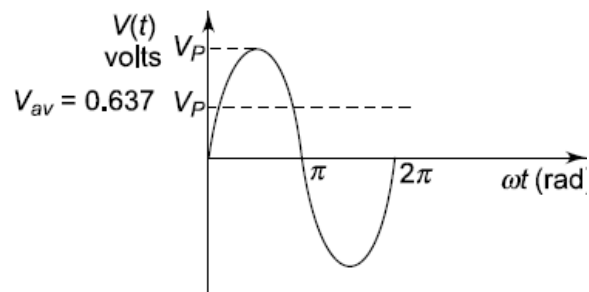


Fig. A 15

Root Mean Square Value or Effective Value

That means the rms value of a sine wave is equal to the dc voltage that produces the same heating effect. In general, the rms value of any function with period T has an effective value given by

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T \overline{v(t)}^2 dt}$$

Consider a function $v(t) = V_P \sin \omega t$

$$\begin{aligned} \text{The rms value, } V_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T (V_P \sin \omega t)^2 d(\omega t)} \\ &= \sqrt{\frac{1}{T} \int_0^{2\pi} V_P^2 \left[\frac{1 - \cos 2\omega t}{2} \right] d(\omega t)} \\ &= \frac{V_P}{\sqrt{2}} = 0.707 V_P \end{aligned}$$

If the function consists of a number of sinusoidal terms, that is

$$\begin{aligned} v(t) &= V_0 + (V_{c1} \cos \omega t + V_{c2} \cos 2 \omega t + \dots) \\ &\quad + (V_{s1} \sin \omega t + V_{s2} \sin 2 \omega t + \dots) \end{aligned}$$

The rms, or effective value is given by

$$V_{\text{rms}} = \sqrt{V_0^2 + \frac{1}{2} (V_{c1}^2 + V_{c2}^2 + \dots) + \frac{1}{2} (V_{s1}^2 + V_{s2}^2 + \dots)}$$

Peak factor :

The peak factor of any waveform is defined as the ratio of the peak value of the wave to the rms value of the wave.

$$\text{Peak factor} = \frac{V_P}{V_{\text{rms}}}$$

$$\text{Peak factor of the sinusoidal waveform} = \frac{V_P}{V_P/\sqrt{2}} = \sqrt{2} = 1.414$$

Form factor :

Form factor of a waveform is defined as the ratio of rms value to the average value of the wave.

$$\text{Form factor} = \frac{V_{\text{rms}}}{V_{\text{av}}}$$

Form factor of a sinusoidal waveform can be found from the above relation.

$$\text{For the sinusoidal wave, the form factor} = \frac{V_P/\sqrt{2}}{0.637 V_P} = 1.11$$

Power factor :

It is the cosine of the angle between voltage and current

Power factor = Active power / Apparent power

5) Explain voltage and current relations in R L and c circuits ?

Ans :

PHASE RELATION IN PURE RESISTOR

When a sinusoidal voltage of certain magnitude is applied to a resistor, a certain amount of sine wave current passes through it. We know the relation between $v(t)$ and $i(t)$ in the case of a resistor. The voltage/current relation in case of a resistor is linear,

i.e.
$$v(t) = i(t) R$$

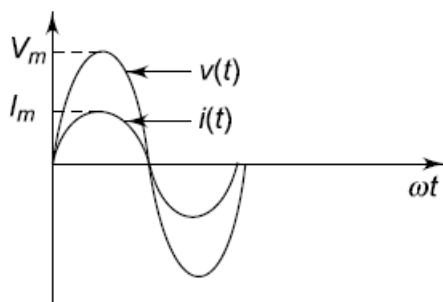
Consider the function

$$i(t) = I_m \sin \omega t = IM [I_m e^{j\omega t}] \text{ or } I_m \angle 0^\circ$$

If we substitute this in the above equation, we have

$$\begin{aligned} v(t) &= I_m R \sin \omega t = V_m \sin \omega t \\ &= IM [V_m e^{j\omega t}] \text{ or } V_m \angle 0^\circ \end{aligned}$$

where
$$V_m = I_m R$$



PHASE RELATION IN A PURE INDUCTOR

$$v(t) = L \frac{di}{dt}$$

Consider the function $i(t) = I_m \sin \omega t = IM [I_m e^{j\omega t}] \text{ or } I_m \angle 0^\circ$

$$v(t) = L \frac{d}{dt} (I_m \sin \omega t)$$

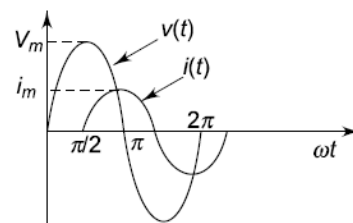
$$= L \omega I_m \cos \omega t = \omega L I_m \cos \omega t$$

$$v(t) = V_m \cos \omega t, \text{ or } V_m \sin (\omega t + 90^\circ)$$

$$= IM [V_m e^{j(\omega t + 90^\circ)}] \text{ or } V_m \angle 90^\circ$$

where $V_m = \omega L I_m = X_L I_m$

and $e^{j90^\circ} = j = 1 \angle 90^\circ$



As a result, in a pure inductor the voltage and current are out of phase. The current lags behind the voltage by 90° in a pure inductor as

PHASE RELATION IN PURE CAPACITOR

As discussed in Chapter 1, the relation between voltage and current is given by

$$v(t) = \frac{1}{C} \int i(t) dt$$

Consider the function $i(t) = I_m \sin \omega t = IM [I_m e^{j\omega t}]$ or $I_m \angle 0^\circ$

$$\begin{aligned} v(t) &= \frac{1}{C} \int I_m \sin \omega t d(t) \\ &= \frac{1}{\omega C} I_m [-\cos \omega t] \end{aligned}$$

$$= \frac{I_m}{\omega C} \sin (\omega t - 90^\circ)$$

$$\begin{aligned} \therefore v(t) &= V_m \sin (\omega t - 90^\circ) \\ &= IM [I_m e^{j(\omega t - 90^\circ)}] \text{ or } V_m \angle -90^\circ \end{aligned}$$

where $V_m = \frac{I_m}{\omega C}$

$$\therefore \frac{V_m \angle -90^\circ}{I_m \angle 0^\circ} = Z = \frac{-j}{\omega C}$$

Hence, the impedance is $Z = \frac{-j}{\omega C} = -jX_C$

where $X_C = \frac{1}{\omega C}$ and is called the capacitive reactance.

If we draw the waveform for both, voltage and current, as shown in Fig. 4.21, there is a phase difference between these two waveforms.

As a result, in a pure capacitor, the current leads the voltage by 90° . The impedance value of a pure capacitor

$$X_C = \frac{1}{\omega C}$$

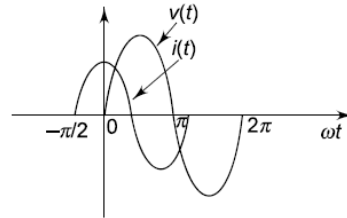


Fig. 4.21